Glaze Search · Quick Start

For new users: install once, then follow one full example, from a photo of a pot to the glaze recipes that could make it. Every screen and number below comes from the app (6 November 2026).

1 · Install once (or use it online)

No install: open `https://huggingface.co/spaces/FBIWarninggg/glaze_search` and log in to Hugging Face (more free GPU time). **On your computer:** clone the branch, then run five launchers in order. On a Mac use the `.command` files (Apple Silicon, M1 or newer).

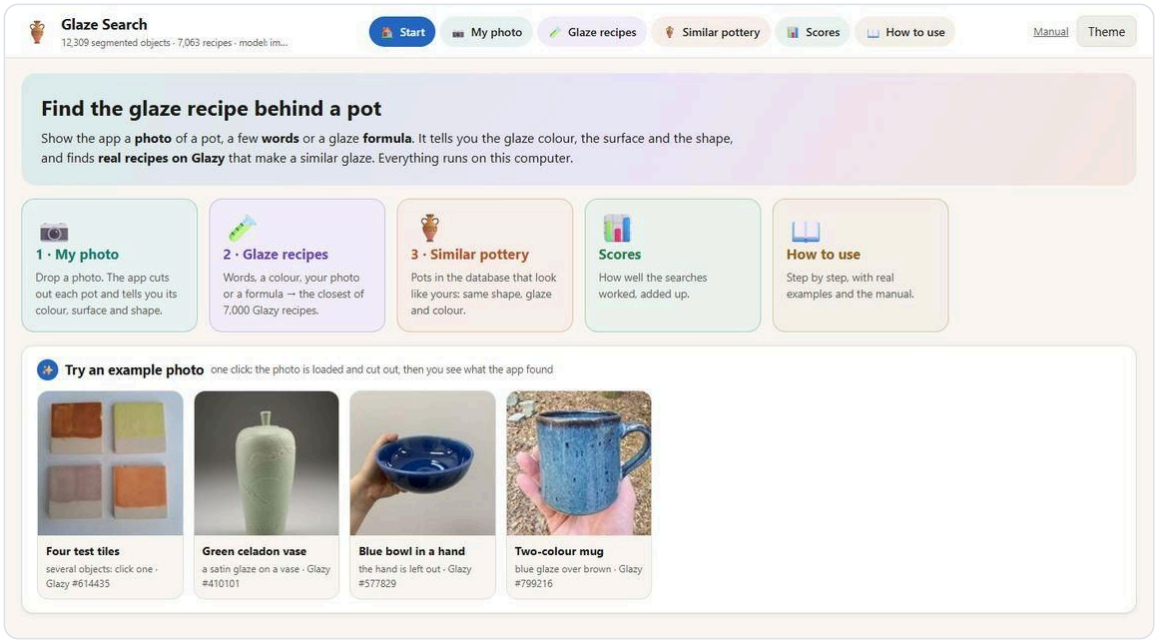
```
git clone -b nghia_mike_ceramic_search --single-branch https://github.com/haohaozai/GC2-Ensemble-Model-Pottery.git
```

#	WINDOWS	MACOS	WHAT IT DOES	WHEN
1	`SETUP.bat`	`SETUP.command`	Python environment and the pretrained models (about 3.7 GB). Needs `uv` installed once.	once
2	`DOWNLOAD_DATA_ORIGINAL_AND_SEGMENTATION` .bat / .command		Photos, cut-outs and search index: 8 archives, about 6.15 GB. Run it again if it stops: it continues.	once
3	`CHECK_DATA.bat`	`CHECK_DATA.command`	Every line should say OK	once
4	`INSTALL_GLAZE_MODEL.bat`	`INSTALL_GLAZE_MODEL.command`	The recipe models from the team's `test_similarity.zip` (drag it onto the .bat; on a Mac put it in Downloads). Only needed for the 🧪 Glaze recipes page.	once
5	`RUN_GLAZE_SEARCH.bat`	`RUN_GLAZE_SEARCH.command`	Starts the app at `http://127.0.0.1:8770` ; keep the window open	every time

2 · The example: which glaze recipe makes this blue bowl?

You have a photo of a glossy blue bowl held in your hand, and you know you fire to cone 6 in an electric kiln (oxidation).

1 Open 📷 My photo



The 🏠 Start page. Click 📷 **My photo** in the bar at the top, or click the example **Blue bowl in a hand** to follow this guide with exactly the same photo.

2 Give it your photo

DO
Drop the photo on box 1, paste it with Ctrl+V (Command+V on a Mac), click to browse, or paste an image link and press **Fetch**. Wait about 5–30 seconds.

YOU SEE
A blue outline around the bowl and "1 object(s) on the photo". Your hand is not outlined: the app recognised it and left it out.



The app's outline of the bowl (object 1). The hand stays outside.



What is searched: the bowl alone (493,207 pixels, 15.7 % of the photo).

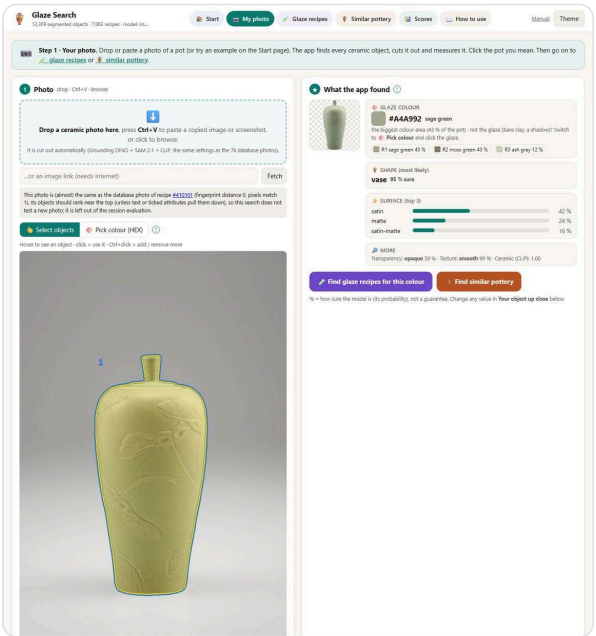
Diagram: A1 Grounding DINO finds it → A2 SAM 2.1 traces the outline → A3 CLIP: ceramic 1.00 ≥ 0.35, keep

3 Read "What the app found"

🎨 GLAZE COLOUR	■ **#1E345B navy blue** · biggest colour area (38 %)
COLOUR REGIONS	navy blue 38 % · denim blue 14 % · blue black 11 % · …
🏺 SHAPE	**bowl** · 97 % sure
✨ SURFACE (TOP 3)	**glossy** · 95 %
🔎 MORE	opaque 50 % · smooth 100 % · ceramic (CLIP) 1.00 · 1 object

Check that the colour is the **glaze**, not bare clay or a shadow. If it is wrong, switch to 🎨 **Pick colour** and click on the glaze, or type a colour in the box (`#1E345B` , `navy blue` , even `dark blu`).

The percentages are how sure the model is, not measured accuracy.



The 📷 My photo page, shown here with another example photo: the photo on the left, **What the app found** on the right.

Diagram: A4 colours (ColorThief + CIELAB) · A5 six attributes (DINO evidence → small trained layer)

4 Press 💉 Find glaze recipes for this colour

DO

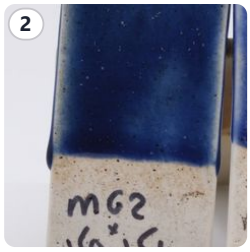
Press the purple button under **What the app found**.

YOU SEE

The 💉 **Glaze recipes** page, already searched with your photo and colour. Only those two parts count, so their weights are photo 54 % · colour 46 %.



Storm Blue - Blue Room Ingredients
View on Glazy ↗ #577829
cone 6 - 8 · Glossy
#1F3153 navy blue · ΔE00 1.7
score **0.912**
photo **0.49**
colour **0.42**



MG2 1Co 1Cu
View on Glazy ↗ #44644
cone 6 · Semi-glossy
#1F2D52 navy blue · ΔE00 3.0
score **0.816**
photo **0.42**
colour **0.40**



12 - Blue haze based on kitten's clear adapt with Gillespie
View on Glazy ↗ #673995
cone 6 · surface ?
#1E3557 navy blue · ΔE00 1.5
score **0.814**
photo **0.39**
colour **0.43**



Floating Blue 3
View on Glazy ↗ #58642
cone 6 · surface ?
#203C5E navy blue · ΔE00 3.5
score **0.790**
photo **0.40**
colour **0.39**



Leach 4321 - Cone 8-9 - Subbing fluxes exercise - RO Sub Wollastonite
View on Glazy ↗ #695243
cone 8 - 9 · Glossy
#27335C navy blue · ΔE00 2.6
score **0.789**
photo **0.38**
colour **0.40**

5 Describe what you want, in your own words

DO

In **1 · Describe it** type

`shinyy dark blu glaze with a few iron specks, cone 6 oxidation`

and press **Read it**. Typos and shorthand are fine.

YOU SEE

What the app understood:

colour dark blue · cone 6 · surface Glossy (from "shinyy") · atmosphere Oxidation · effect speckled (from "specks")

Typos fixed: shinyy → shiny · blu → blue · not used: few, iron.

The colour picked on the photo stays in use: it is more exact than the words "dark blue". Grey suggestion chips (for example "family: brown") are only hints and are never scored.

Diagram: B3 read the words (rules + fuzzy matching) · B3 suggest tags (TF-IDF) · B2 MiniLM turns the sentence into 384 numbers

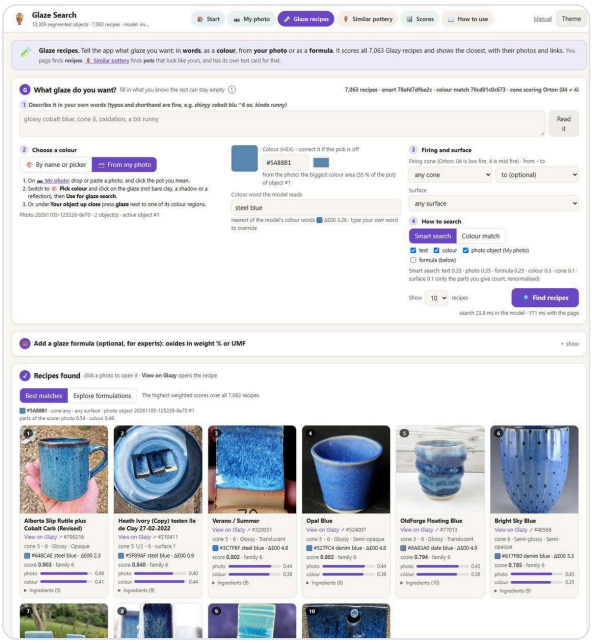
6 Check the firing and the engine, then search

DO

3 · Firing and surface: cone 6 and Glossy are filled in from your words (cone **04** is low fire, not cone 4). **4 · How to search**: **Smart search**. **Show 10**. Press 🔍 **Find recipes**.

WHAT COUNTS

The parts you gave, rescaled to add up to 1: words 0.28 · photo 0.28 · colour 0.24 · cone 0.08 · surface 0.08 · atmosphere 0.04. It takes under a second.



The 🧪 Glaze recipes page: 1 · Describe it, 2 · the colour, 3 · firing and surface, 4 · how to search, then the results (shown with another example).

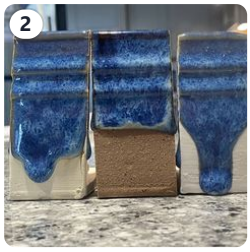
Diagram: join → C2 smart search (words + photo + formula heads on frozen MiniLM and DINOv2) → rank every recipe

7 Read the result: Best matches



Storm Blue - Blue Room Ingredients
View on Glazy ↗ #577829
cone 6 - 8 · Glossy
#1F3153 navy blue · ΔE00 1.7
score **0.847** ✓ cone ✓ surface

words	0.17
photo	0.25
colour	0.22
cone	0.08
surface	0.08
atmosphere	0.04



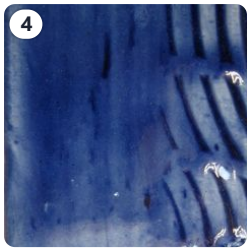
Louck's Floating Blue
View on Glazy ↗ #64308
cone 6 · Glossy
#3A4D6B navy blue · ΔE00 3.6
score **0.798** ✓ cone ✓ surface

words	0.20
photo	0.20
colour	0.20
cone	0.08
surface	0.08
atmosphere	0.04



Cloudy night
View on Glazy ↗ #714462
cone 6 · Glossy
#1D3047 navy blue · ΔE00 5.3
score **0.786** ✓ cone ✓ surface

words	0.19
photo	0.21
colour	0.18
cone	0.08
surface	0.08
atmosphere	0.04



Dolita Base - Dark Blue
View on Glazy ↗ #3118
cone 6 · Glossy
#2A4075 navy blue · ΔE00 5.0
score **0.778** ✓ cone ✓ surface

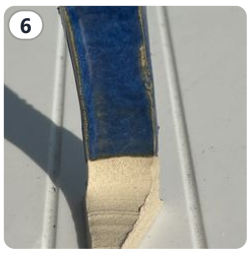
words	0.19
photo	0.20
colour	0.19
cone	0.08
surface	0.08
atmosphere	0.04



Clear Blue Glaze (No EPK) B
View on Glazy ↗ #688871
cone 6 · Glossy
#1A3A6D navy blue · ΔE00 3.4
score **0.777** ✓ cone ✓ surface

words	0.18
photo	0.20
colour	0.20
cone	0.08
surface	0.08
atmosphere	0.04

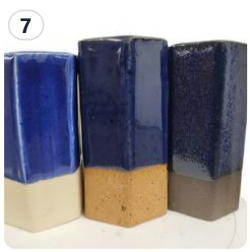
6



Floating Blue with MnCO3
View on Glazy ↗ #660989
cone 6 · Glossy
#283C54 navy blue · ΔE00 5.5
score **0.775** ✓ cone ✓ surface

words	0.19
photo	0.21
colour	0.18
cone	0.08
surface	0.08
atmosphere	0.04

7



Spruill #22 Cobalt
View on Glazy ↗ #758437
cone 5 - 6 · Glossy
#313C78 navy blue · ΔE00 5.6
score **0.775** ✓ cone ✓ surface

words	0.18
photo	0.21
colour	0.18
cone	0.08
surface	0.08
atmosphere	0.04

8



DEEP OCEAN
View on Glazy ↗ #690411
cone 6 · Glossy
#26384B navy blue · ΔE00 7.2
score **0.773** ✓ cone ✓ surface

words	0.19
photo	0.22
colour	0.17
cone	0.08
surface	0.08
atmosphere	0.04

9



French Blue
View on Glazy ↗ #727202
cone 5 - 6 · Glossy
#243342 blue black · ΔE00 8.8
score **0.772** ✓ cone ✓ surface

words	0.20
photo	0.22
colour	0.15
cone	0.08
surface	0.08
atmosphere	0.04

10



Runny Blue
View on Glazy ↗ #73100
cone 6 · Glossy
#324866 navy blue · ΔE00 7.0
score **0.772** ✓ cone ✓ surface

words	0.21
photo	0.19
colour	0.17
cone	0.08
surface	0.08
atmosphere	0.04

How to read a card. The Glazy photo and name (View on Glazy opens the recipe); cone and surface, with ✓ when they match yours; the colour with its **ΔE00** (below about 5 the colours look alike, above 10 they are different colours); the score, with bars showing what each part added. For #1: words 0.17 + photo 0.25 + colour 0.22 + cone 0.08 + surface 0.08 + atmosphere 0.04 = **0.847**.

8 Explore formulations: other chemistries for a similar look

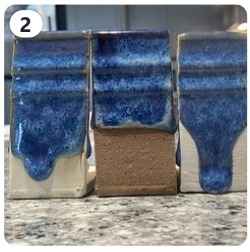
Press the **Explore formulations** tab: at most 2 recipes per chemistry family, so you see different bases, not ten copies of one.

1



Storm Blue - Blue Room Ingredients
View on Glazy ↗ #577829
cone 6 - 8 · Glossy
#1F3153 navy blue · ΔE00 1.7
score **0.847** ✓ cone ✓ surface

2



Louck's Floating Blue
View on Glazy ↗ #64308
cone 6 · Glossy
#3A4D6B navy blue · ΔE00 3.6
score **0.798** ✓ cone ✓ surface

3



June Perry Aquamarine
View on Glazy ↗ #209746
cone 5 - 7 · Glossy
#23879F cerulean blue · ΔE00 3.5
score **0.744** ✓ cone ✓ surface

4



OldForge Misty Forest (FirstFive)
View on Glazy ↗ #228056
cone 5 - 7 · Glossy
#4C6C5A forest green · ΔE00 4.1
score **0.720** ✓ cone ✓ surface

5



Recette 9 : Goutte d'huile
View on Glazy ↗ #295891
cone 6 - 8 · Glossy
#2C333A charcoal black · ΔE00 12.7
score **0.720** ✓ cone ✓ surface

9 (Optional) Ask the second engine: Colour match

Switch **4 · How to search** to **Colour match** and press 🔍 **Find recipes**. It reads only colour + cone + surface (as `color: navy blue, Cone 6, Glossy, unknown, unknown`) and compares them with each recipe's chemistry: 0.40 learned match + 0.40 colour + 0.10 surface + 0.10 cone.



Storm Blue - Blue Room Ingredients
View on Glazy ↗ #577829
cone 6 - 8 · Glossy
#1F3153 navy blue · ΔE00 1.7
score **0.876** ✓ cone ✓ surface

learned match	0.31
colour	0.37
surface	0.10
cone	0.10



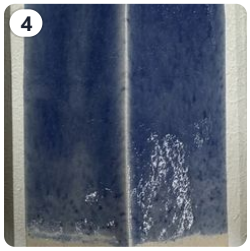
Studio Neves Azul GA42
View on Glazy ↗ #97643
cone 5 1/2 - 6 · Glossy
#223761 navy blue · ΔE00 1.3
score **0.793** ✓ cone ✓ surface

learned match	0.22
colour	0.38
surface	0.10
cone	0.10



Clear Blue Glaze (No Neph Sy) A
View on Glazy ↗ #696210
cone 6 · Glossy
#2E3654 navy blue · ΔE00 4.5
score **0.776** ✓ cone ✓ surface

learned match	0.26
colour	0.32
surface	0.10
cone	0.10



Blue Jeans Candy coat
View on Glazy ↗ #631653
cone 6 - 8 · Glossy
#36425C navy blue · ΔE00 6.3
score **0.773** ✓ cone ✓ surface

learned match	0.28
colour	0.29
surface	0.10
cone	0.10



Floating Blue (Milk Oolong Pottery)
View on Glazy ↗ #542632
cone 5 - 6 · Glossy
#1A406B navy blue · ΔE00 5.0
score **0.765** ✓ cone ✓ surface

learned match	0.25
colour	0.31
surface	0.10
cone	0.10

When both engines put the same recipe first, you can trust it more.

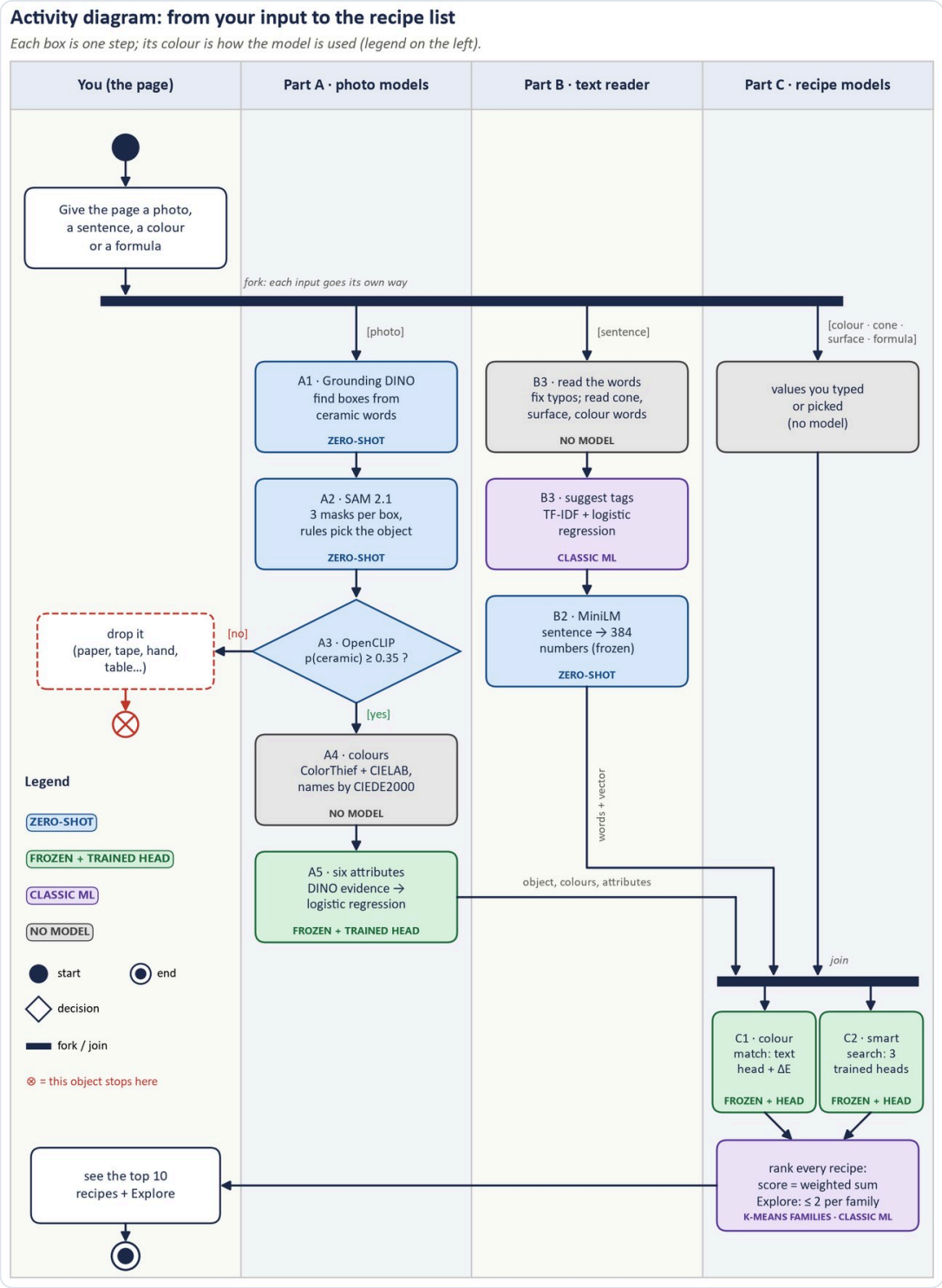
10 Decide

Open the top cards with **View on Glazy ↗** to read their ingredients. Prefer a small ΔE00 and ✓ cone for your kiln, then **test-fire a small tile** before you mix a big batch.

The answer for this bowl: Storm Blue - Blue Room Ingredients (Glazy #577829, cone 6 - 8, glossy, ΔE00 1.7): first in both engines.
It is in fact this bowl's own recipe: the example photo comes from the Glazy database, so the app found the exact recipe from a photo and a sentence with typos. With a photo of your own pot there is no "exact" recipe: the list gives the closest glazes (here other glossy cone-6 navy blues such as Louck's Floating Blue and Cloudy night), and you choose with ΔE00 and ✓ cone.

3 · What happened behind each step

STEP	BOX IN THE ACTIVITY DIAGRAM	HOW THE MODEL IS USED
2	A1 Grounding DINO · A2 SAM 2.1 · A3 CLIP	finds, outlines and checks the pot (zero-shot)
3	A4 colours · A5 six attributes	ColorThief + CIELAB; DINO evidence → small trained layer
5	B3 read the words · B3 suggest tags · B2 MiniLM	rules + fuzzy matching; TF-IDF hints; frozen text model
6	values you typed or picked → join	colour, cone, surface (and a formula, if you give one)
6–7	C2 smart search (3 trained heads) → rank	words + photo + colour + cone + surface + atmosphere
9	C1 colour match (text ↔ chemistry)	colour + cone + surface
8	Explore: ≤ 2 per K-Means family	different chemistries for a similar look



The activity diagram of one search: the fork sends each input to its own lane, the join collects everything for the recipe models.

4 · Problems and where to read more

IF…	THEN…
the colour is the bare clay or a shadow	🎨 Pick colour and click on the glaze, or type the colour in the box
the wrong pot is used	click the pot you mean (yellow outline = the one in use)
online: “Hugging Face's free GPU: RuntimeError”	try again in a minute; the Space owner can restart the Space (⋮ → Restart Space)
online: “free GPU limit”	log in to Hugging Face, or wait: the allowance renews
“the formula is not used”	give at least 3 oxides, including a flux (CaO, K2O, Na2O, MgO, ZnO …)

DOCUMENT (DOCS/MANUALS)	WHAT IS IN IT
CG2_Glaze_Search_Manual (.html / .pdf)	all six pages and six worked examples, every number explained
CG2_Glaze_Search_Process_Models_MindMap.pdf	the whole pipeline, the 10 steps and their settings
CG2_Glaze_Search_Pretrained_Models.pdf	how each pretrained model works, with figures
CG2_Glaze_Search_Evaluation.pdf	every metric and test result, stage by stage